

Haoyang He (Alan)

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PROFILE

Quantitative STEM graduate moving into credit risk. I analyse large, multi-source datasets in SQL and Python, build and calibrate quantitative models, and spot the trends and root-cause drivers behind a change in performance. I turn that analysis into clear reporting and recommendations for stakeholders, and I want to apply it to portfolio monitoring and lending decisions within risk appetite.

EDUCATION

MSc Advanced Materials Science, University College London 09/2025 – 09/2026
Predicted Distinction. Research project on data-driven device characterisation and model fitting.

BSc (Hons) Physics, University of Edinburgh 09/2021 – 05/2025
Quantitative modules: Statistics, Computer Modelling, Thermodynamics, Quantum Physics, Electromagnetism, Condensed Matter Physics.

EXPERIENCE

Risk Job Simulation, Goldman Sachs 09/2026

- Completed a job simulation involving risk assessment and management for Goldman Sachs' Risk team.
- Conducted individual credit risk analysis using quantified financial data, enhancing skills in personal finance evaluation.
- Applied critical thinking and quantitative analysis to real estate scenarios, interpreting data to assess financial implications.
- Developed and recommended risk mitigation strategies, demonstrating practical application of financial principles and market understanding.

Semiconductor Device Data Analysis Project, UCL 11/2025 – 08/2026

- Build quantitative models in Python (nonlinear regression) to fit time-dependent relaxation data, extract key parameters and benchmark performance across devices.
- Separate slow ionic drift from fast electronic effects in photocurrent data, attributing each change in performance to its root-cause driver rather than to noise.

Sales Performance & Customer Segmentation Analysis, Independent Project 09/2025

- Queried 10,000+ transaction records in SQL (CTEs, window functions, aggregations) to track year-on-year revenue, product performance and seasonality, identifying the trends behind demand shifts.
- Segmented the customer portfolio with RFM and Pareto analysis to rank value and concentration, flagging high-value cohorts for retention.
- Turned findings into concise, management-information style insight to support revenue, inventory and retention decisions.

Biofilm Mechanics Research Project, University of Edinburgh 01/2025 – 04/2025

- Designed and led a 10-week study comparing four gene-deletion mutants with the wild type, using amplitude-sweep stress tests to find the strain limit at which each material fails.
- Quantified impact precisely: removing the TasA gene cut stiffness (storage modulus) by over 75% and raised the failure threshold five-fold.
- Built a reproducible Python workflow (data cleaning, validation, interpolation, Welch's t-tests) that cut manual analysis time and gave statistically robust comparisons across 30+ datasheets.

Fluid Mechanics Research Intern, University of Edinburgh 06/2024 – 08/2024

- Identified the critical threshold (Reynolds number) at which flow shifts regime, by systematically varying fluid viscosity in rotational rheometry.
- Extended a published 1985 model to non-Newtonian fluids and built a Python nonlinear regression model that narrowed the parameter range describing the transition.

Shell Engineer, HYPED Hyperloop Team, University of Edinburgh 09/2022 – 07/2023

- Owned aerodynamic design and manufacture of the pod shell for the UK's first student-built Hyperloop test pod, aligning design decisions with cross-functional team members.
- Weighed cost against performance by choosing fibreglass over carbon fibre, cutting shell cost by about 20% while maintaining the test performance.

SKILLS

- Data & programming:** SQL (MySQL: CTEs, window functions, aggregations), Python (pandas, NumPy, scikit-learn), R
- Modelling & statistics:** linear and nonlinear regression, classification, decision trees
- Analysis & reporting:** trend and seasonality analysis, cohort and RFM segmentation, Excel, Power BI, Tableau
- Certificates:** Business Analytics (Wharton), Supervised Machine Learning (Stanford), Failure Mode and Effects Analysis (CPD)
- Languages:** English (fluent), Mandarin (native), German (Cambridge Pre-U Distinction)
- Interests:** Metal band drummer; Tour du Mont Blanc finisher; Kilimanjaro summitter